Sodium bicarbonate: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Sodium bicarbonate: Patient drug information" and "Sodium bicarbonate: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Pharmacologic Category

- Alkalinizing Agent;
- Antacid;
- Electrolyte Supplement, Oral;
- Electrolyte Supplement, Parenteral

Dosing: Adult

Dosage guidance:

Safety: Avoid extravasation; tissue necrosis may occur due to hypertonicity and alkalinity (Ref).

Dosage form information: Each 650 mg oral tablet contains ~7.7 mEq of sodium and 7.7 mEq of bicarbonate. Each ½ teaspoon of baking soda contains ~27 mEq of sodium and 27 mEq bicarbonate (may vary slightly depending on product and accuracy of measuring device) (Ref). Each mL of 8.4% IV product provides 1 mEq/mL of sodium and 1 mEq/mL of bicarbonate.

Antacid

Antacid: Oral: 650 mg to 2.6 g every 4 hours as needed; maximum daily dose: 15.6 g/day. Do not use maximum dosage for >2 weeks.

Cardiac conduction delays due to sodium channel blockade

Cardiac conduction delays due to sodium channel blockade (off-label use): Note: The optimal dose has not been established (Ref). Also, the optimal QRS threshold at which to initiate therapy has not been established; some experts recommend initiating therapy if the QRS interval is >100 msec (Ref), whereas others state a threshold of >120 msec (Ref). Consultation with a clinical toxicologist or poison center is highly recommended.

Initial : IV: 50 to 100 mEq **or** 1 to 2 mEq/**kg** over 1 to 2 minutes; repeat as needed to achieve narrowing of the QRS interval while maintaining a blood pH of 7.45 to 7.55. May initiate a continuous infusion following initial bolus therapy (Ref).

Continuous infusion: IV: Following initial bolus therapy, initiate a continuous infusion using sodium bicarbonate (8.4%) 150 mEq in 1 L D5W. Administer at a rate of ~150 mL/hour to maintain a serum pH of 7.45 to 7.55. Avoid fluid overload (Ref).

Hyperkalemia, severe/emergent

Hyperkalemia , severe/emergent (adjunctive agent) (off-label use): Note: Practice may vary; refer to institutional protocols.

Hyperkalemia with cardiac arrest: Intermittent bolus: **IV:** 50 mEq over 5 minutes (Ref).

Hyperkalemia with metabolic acidosis: Note: May consider in patients with persistent severe hyperkalemia and/or ECG changes despite calcium and other therapies to decrease serum potassium, particularly in patients with metabolic acidosis (Ref).

IV: 150 mEq in 1 L of D5W over 2 to 4 hours (Ref).

Metabolic acidosis, acute severe

Metabolic acidosis, acute severe: Note: Must treat underlying cause; the underlying cause and degree of acidosis may result in the need for larger or smaller bicarbonate replacement doses. Consider bicarbonate therapy in patients with either a pH of <7.1 **or** in patients with severe acute kidney injury and a pH of ≤7.2. In most cases, the initial goal of therapy is to maintain a target pH >7.2 until the primary process causing metabolic acidosis is resolved. In patients with a pH of ≤7.2 and severe acute kidney injury, the initial pH goal is >7.3 (Ref). Optimal dosing, regimens, administration, and monitoring have not been identified; refer to institutional protocols.

Example regimens:

Intermittent therapy: 7.5% or 8.4% sodium bicarbonate: **IV:** Initial: 89.2 to 100 mEq once over 1 to 2 minutes; reassess pH, serum bicarbonate level, and clinical status every 2 hours. If pH remains below target (~7.2 to 7.3), administer an additional 44.6 to 100 mEq sodium bicarbonate or initiate a continuous infusion (Ref).

Example: 100 mEq of 8.4% sodium bicarbonate once over 1 to 2 minutes, then repeat with 50 to 100 mEq or initiate a continuous infusion if pH remains below target (Ref).

Continuous infusion therapy:

Bicarbonate deficit regimen: Note: Dosing is based on the following bicarbonate deficit formula. This equation provides a very rough estimate of the initial bicarbonate replacement dose; monitor clinical status closely. Some experts prefer intermittent bolus therapy over continuous infusion (Ref).

Continuous infusion: **IV:** Sodium bicarbonate estimated dose (mEq) = 0.5 × weight (kg) × [goal serum bicarbonate – observed serum bicarbonate (mEq/L)]; generally, goal serum bicarbonate is ~8 to 12 mEq/L (Ref). **Note:** In the equation above, ”0.5 × weight (kg)” represents the estimated bicarbonate V_d.

Note: Administer the calculated amount of bicarbonate (mEq) over several hours (eg, 2 to 4 hours) until pH is ~7.2 to 7.3; reassess pH, serum bicarbonate level, and clinical status every 2 hours, and adjust dose as needed until goals are reached (Ref).

Example: For estimated bicarbonate deficit of 150 mEq, administer sodium bicarbonate [8.4%] 150 mEq in 1 L D5W over 2 to 4 hours, then reassess pH, serum bicarbonate level, and clinical status. Adjust dose if pH remains below target (Ref).

Metabolic acidosis in patients with chronic kidney disease

Metabolic acidosis in patients with chronic kidney disease (off-label dose): Note: Kidney Disease Improving Global Outcomes guidelines suggest oral replacement when plasma HCO_3^- concentrations are <22 mEq/L (Ref).

Oral: Initial: 15.4 to 23.1 mEq/day in divided doses (eg, 650 mg tablet 2 to 3 times daily); titrate to normal serum bicarbonate concentrations (eg, 23 to 29 mEq/L, although a more targeted range of 24 to 26 mEq/L has been suggested by observational studies (Ref)) or up to 5,850 mg/day; sodium bicarbonate powder may be used as an alternative in patients who cannot take tablets (Ref). Avoid exceeding serum bicarbonate concentrations >29 mEq/L since this has been associated with increased mortality in patients with chronic kidney disease (Ref).

Neutralize lidocaine with epinephrine dental anesthetic

Neutralize lidocaine with epinephrine dental anesthetic: Neutralizing additive: Mix 10 parts anesthetic (lidocaine with epinephrine) to 1 part 8.4% sodium bicarbonate.

Add 0.18 mL sodium bicarbonate to 1.8 mL cartridge of lidocaine 2% with epinephrine 1:50,000 or 1:100,000.

Add 2 mL sodium bicarbonate to 20 mL vial of lidocaine 2% with epinephrine 1:100,000.

Add 3 mL sodium bicarbonate to 30 mL vial of lidocaine 2% with epinephrine 1:100,000.

Add 5 mL sodium bicarbonate to 50 mL vial of lidocaine 2% with epinephrine 1:100,000.

Prevention of contrast-induced nephropathy

Prevention of contrast-induced nephropathy (off-label use): IV infusion: 154 mEq/L sodium bicarbonate in D5W solution: 3 mL/kg/hour for 1 hour immediately before contrast injection, then 1mL/kg/hour during contrast exposure and for 6 hours after procedure (Ref). Some have described a prophylactic strategy based on patient risk for contrast induced nephropathy and procedure type (Ref). **Note:** In patients at high risk for renal complications, no benefit was seen with IV sodium bicarbonate over IV sodium chloride in one study; therefore, some consider IV sodium chloride as the preferred option due to lower cost and no need for compounding (Ref).

Renal tubular acidosis

Renal tubular acidosis:

Distal (off-label dose): **Oral**: Initial: 1 to 2 mEq/kg/day in up to 4 divided doses; titrate as necessary to maintain serum bicarbonate in the target range (Ref).

Proximal (off-label dose): **Oral:** Initial: 10 to 15 mEq/kg/day in up to 4 divided doses; titrate as necessary to maintain serum bicarbonate in the target range (Ref). **Note:** Large, infrequent doses may lead to excess bicarbonate loss in the urine; hence smaller, more frequent dosing is preferred (Ref).

Urine alkalinization

Urine alkalinization (off-label dose):

Oral: Initial: 48 mEq (4 g), then 12 to 24 mEq (1 to 2 g) every 4 hours; dose should be titrated to desired urinary pH; doses up to 186 mEq/day (15.6 g/day) in patients <60 years of age and 92.9 mEq/day (7.8 g/day) in patients >60 years of age. Administration of 48 mEq (4 g) every 8 hours has also been shown to achieve a urinary pH of at least 7 after a period of 10 hours in one study of healthy volunteers (Ref).

Urine alkalinization for high-dose methotrexate: **Oral:** 1.3 g every 4 hours while awake beginning the morning of scheduled admission day, followed by IV hydration containing sodium bicarbonate after admission (Ref) **or** 1.3 to 1.5 g every 6 hours (with IV hydration with Lactated Ringer's) beginning in the evening prior to high-dose methotrexate administration; if the urine pH failed to reach the target (pH ≥7), the dose could be escalated to 6.5 g every 6 hours, and further escalated up to 10 g every 6 hours (Ref). Sodium bicarbonate has also been used in combination with acetazolamide to facilitate urine alkalization in the ambulatory setting or in instances of IV sodium bicarbonate shortages (Ref).

IV (off label): Note: IV administration is recommended for the treatment of specific toxicities (eg, salicylates) (Ref); consultation with a clinical toxicologist or poison center is highly recommended.

Initial: Note: May consider use of bolus doses prior to initiation of a continuous infusion, especially in patients with preexisting acidemia (Ref).

IV: 50 to 100 mEq or 1 to 2 mEq/kg over 1 to 2 minutes; repeat as needed to achieve a urinary pH of 7.5 to 8.5 and a serum pH of 7.45 to 7.55 (Ref). Initiate a continuous infusion following bolus dose(s) (if used) (Ref).

Continuous infusion: IV: Initiate a continuous infusion using sodium bicarbonate (8.4%) 150 mEq in 1 L D5W. Administer at a rate of ~200 to 250 mL/hour to maintain a urinary pH of 7.5 to 8.5 and a serum pH of 7.45 to 7.55. Avoid fluid overload (Ref).

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Contains sodium; use with caution, especially in patients with concomitant hypertension, heart failure, or volume overload (Ref).

Altered kidney function: No dosage adjustment necessary for any degree of kidney dysfunction (Ref).

Hemodialysis, intermittent (thrice weekly): No dosage adjustment necessary (Ref).

Peritoneal dialysis: No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution, especially in clinical states associated with edema and sodium retention.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Sodium bicarbonate: Pediatric drug information")

Dosage guidance:

Dosing: Dose should be individualized to patient response and target parameters for condition being treated.

Antacid

Antacid: Note: Chronic antacid therapy not recommended for management of gastroesophageal reflux disease in pediatric patients (Ref).

Children ≥6 years and Adolescents: Powder for solution: Oral: $^{1}/_{2}$ teaspoonful in 4 ounces of water/dose; may repeat up to every 2 hours; maximum daily dose: 6 doses/**day**; do not use maximum dose for longer than 2 weeks.

Cardiac arrest

Cardiac arrest (PALS guidelines): Limited data available:

Note: Commercially available products: Each mL of 4.2% IV product provides 0.5 mEq/mL of sodium and 0.5 mEq/mL of bicarbonate. Each mL of 8.4% IV product provides 1 mEq/mL of sodium and 1 mEq/mL of bicarbonate.

Infants, Children, and Adolescents: IV, Intraosseous: 1 mEq/kg/dose; repeat doses should be guided by arterial blood gases; per the manufacturer, the 4.2% (0.5 mEq/mL) solution may be preferred in infants and children <2 years of age. **Note:** If intraosseous route is used for administration and is subsequently used to obtain blood samples for acid-base analysis, results will be inaccurate (Ref).

Note: Routine use of sodium bicarbonate ($NaHCO_3$) in cardiac arrest is not recommended. May be considered in the setting of prolonged cardiac arrest only after adequate alveolar ventilation has been established and effective cardiac compressions. Sodium bicarbonate may be indicated in some cardiac arrest situations (eg, hyperkalemia, sodium channel blocker toxicity [eg, tricyclic antidepressant overdose]) (Ref).

Chronic kidney disease acidosis

Chronic kidney disease (CKD) acidosis: Limited data available: **Note:** Initiate if serum bicarbonate <22 mEq/L:

Infants, Children, and Adolescents: Oral: Initial dose based on serum bicarbonate levels (see following equation); calculate the deficit to determine dose needed; may divide dose for tolerability; adjust dose to maintain serum bicarbonate within the targeted normal range (eg, children: 22 to 23 mEq/L; adults: 24 to 25 mEq/L); undertreatment should be avoided due to negative effects of acidosis on growth (Ref).

HCO_3^- (mEq) deficit = 0.5 × weight (kg) × [desired HCO_3^- (mEq/L) − measured serum HCO_3^- (mEq/L)]

Hyperkalemia; adjunct

Hyperkalemia; adjunct: Limited data available; efficacy results variable:

Note: Commercially available products: Each mL of 4.2% IV product provides 0.5 mEq/mL of sodium and 0.5 mEq/mL of bicarbonate. Each mL of 8.4% IV product provides 1 mEq/mL of sodium and 1 mEq/mL of bicarbonate.

Infants, Children, and Adolescents: IV or Intraosseous: 1 to 2 mEq/kg/dose (maximum dose: 50 mEq/dose) has been used to redistribute extracellular potassium into cells; not recommended for routine use for cardiac arrest (Ref).

Metabolic acidosis, acute

Metabolic acidosis, acute: Limited data available:

Note: Commercially available products: Each mL of 4.2% IV product provides 0.5 mEq/mL of sodium and 0.5 mEq/mL of bicarbonate. Each mL of 8.4% IV product provides 1 mEq/mL of sodium and 1 mEq/mL of bicarbonate.

Blood-gas directed dosing (equations): Infants, Children, and Adolescents: IV: These equations provide an estimated replacement dose. The underlying cause and degree of acidosis may result in the need for larger or smaller replacement doses. In most cases, the initial goal of therapy is to target a pH of ~7.2 to prevent overalkalinization (Ref).

HCO_3^- (mEq) deficit = 0.3 × weight (kg) × base deficit (mEq/L) **or**

HCO_3^- (mEq) deficit = 0.3 to 0.5 × weight (kg) × [desired HCO_3^- (mEq/L) − measured serum HCO_3^- (mEq/L)]

Weight-directed dosing (if acid-base status is not available): Infants, Children, and Adolescents: IV, Intraosseous: 0.5 to 1 mEq/kg/dose over 5 to 15 minutes; maximum dose: 50 mEq/dose (Ref). Subsequent doses should be based on patient's acid-base status.

Neutralization local anesthetic, dental

Neutralization local anesthetic (lidocaine with epinephrine), dental:

Children and Adolescents: Neutralizing additive: Mix 10 parts anesthetic (lidocaine with epinephrine) to 1 part 8.4% sodium bicarbonate.

Add 0.18 mL 8.4% sodium bicarbonate to 1.8 mL cartridge of lidocaine 2% with epinephrine 1:50,000 **or** 1:100,000.

Add 2 mL 8.4% sodium bicarbonate to 20 mL vial of lidocaine 2% with epinephrine 1:100,000.

Add 3 mL 8.4% sodium bicarbonate to 30 mL vial of lidocaine 2% with epinephrine 1:100,000.

Add 5 mL 8.4% sodium bicarbonate to 50 mL vial of lidocaine 2% with epinephrine 1:100,000.

Overdose, sodium channel blocker

Overdose, sodium channel blocker (eg, tricyclic antidepressant [TCA]): Limited data available:

Note: Commercially available products: Each mL of 4.2% IV product provides 0.5 mEq/mL of sodium and 0.5 mEq/mL of bicarbonate. Each mL of 8.4% IV product provides 1 mEq/mL of sodium and 1 mEq/mL of bicarbonate.

Infants, Children, and Adolescents: IV: 1 to 2 mEq/kg/dose; may repeat in 5 minutes if no response, followed by continuous IV infusion of 150 mEq $NaHCO_3$/L solution to maintain targeted pH. **Note:** Goal pH of 7.5 to 7.55 is recommended in TCA poisoning with hypotension, widened QRS interval >100 ms, or ventricular arrhythmia (Ref).

Renal tubular acidosis

Renal tubular acidosis (RTA): Limited data available: **Note:** Dose should be individualized based on urinary bicarbonate excretion (degree depends on type of RTA), serum bicarbonate, and possibly age-related factors.

Infants, Children, and Adolescents:

Distal; type 1: Oral: Usual dose: 3 to 4 mEq bicarbonate/kg/**day** in divided doses; titrate to maintain normal electrolyte concentrations; younger patients require more bicarbonate replacement due to acid production during growth; some data suggest infants require a higher dose of 5 to 10 mEq bicarbonate/kg/**day** and growing children need 4 to 8 mEq bicarbonate/kg/**day** (Ref).

Proximal, type 2: Oral: Usual dose: 10 to 15 mEq/kg/**day** in divided doses; titrate as necessary to maintain normal electrolyte concentrations; doses as high as 20 mEq bicarbonate/kg/**day** have been described (Ref).

Skin protectant; relief of minor irritation

Skin protectant; relief of minor irritation: Note: Notify physician if no symptom resolution within 14 days, or if symptoms reappear after an initial resolution.

Children ≥2 years and Adolescents: Topical: Powder for solution (baking soda):

Bath soak: Topical: Dissolve 1 to 2 cups of powder in tub of warm water and soak for 10 to 30 minutes; pat skin dry (do not rub).

Compress or wet dressing: Topical: Mix powder with water; soak clean, soft cloth in mixture and apply cloth loosely to affected area for 15 to 30 minutes; may repeat as needed or as directed by physician.

Paste: Topical: Mix powder with water to form a paste and apply to the affected area of skin as needed.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution; can cause sodium retention.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution, especially in clinical states associated with edema and sodium retention.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined: Endocrine & metabolic: Hypernatremia, metabolic alkalosis

Contraindications

Injection: Chloride loss due to vomiting or from continuous GI suction; concomitant use of diuretics known to produce a hypochloremic alkalosis.

Neutralizing additive (dental or IV use): Not for use as a systemic alkalizer.

OTC labeling: When used for self-medication, do not use if on low-sodium diet.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

• Extravasation: Vesicant (at concentrations ≥8.4%); ensure proper catheter or needle position prior to and during infusion. Avoid extravasation (tissue necrosis may occur due to hypertonicity and alkalinity) (Ong 2020).

Disease-related concerns:

• Cirrhosis: Use with caution in patients with cirrhosis.

• Edema: Use with caution in patients with edema.

• Heart failure: Use with caution in patients with heart failure.

• Peptic ulcer disease: Not to be used in treatment of peptic ulcer disease.

• Kidney impairment: Use with caution in patients with kidney impairment; may cause sodium retention.

Special populations:

• Older adult: Not the antacid of choice for the older adults because of sodium content and potential for systemic alkalosis.

• Pediatric: Rapid administration in neonates, infants, and children <2 years of age has led to hypernatremia, decreased CSF pressure, and intracranial hemorrhage.

Dosage form specific issues:

• Injection: Use of IV sodium bicarbonate should be reserved for documented severe metabolic acidosis and for severe/emergent hyperkalemia (eg, cardiotoxicity).

• Powder/Tablets: Completely dissolve in water prior to administration; severe stomach irritation may occur.

Other warnings/precautions:

• Self-medication (OTC use): Antacid: When used for self-medication (OTC), ask health care provider prior to use if on a sodium-restricted diet. Discontinue use and notify health care provider if the maximum dosage has been used for 2 weeks or severe abdominal pain occurs with use. Do not administer when overly full from food or drink.

Dosage Forms Considerations

Sodium bicarbonate solution 4.2% [42 mg/mL] provides 0.5 mEq/mL each of sodium and bicarbonate

Sodium bicarbonate solution 7.5% [75 mg/mL] provides 0.9 mEq/mL each of sodium and bicarbonate

Sodium bicarbonate solution 8.4% [84 mg/mL] provides 1 mEq/mL each of sodium and bicarbonate

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Powder, Oral:

Generic: (1 g [DSC], 113 g, 454 g, 500 g, 1000 g, 2500 g, 10000 g, 12000 g, 24000 g, 25000 g, 45000 g)

Solution, Intravenous:

Generic: 4.2% (10 mL); 8.4% (10 mL, 50 mL)

Solution, Intravenous [preservative free]:

Generic: 4.2% (5 mL); 7.5% (50 mL [DSC]); 8.4% (50 mL)

Tablet, Oral:

Generic: 325 mg, 650 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Sodium Bicarbonate Intravenous)

4.2% (per mL): $1.06 - $5.65

8.4% (per mL): $0.15 - $0.60

Tablets (Sodium Bicarbonate Oral)

325 mg (per each): $0.01 - $0.05

650 mg (per each): $0.01 - $0.05

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous:

Generic: 4.2% (10 mL); 7.5% ([DSC]); 8.4% (10 mL, 50 mL)

Administration: Adult

IV:

Direct IV injection: No further dilution required; administer by rapid IV push (Ref); refer to institutional policies and procedures.

IV infusion: Dilute to a maximum concentration of 0.5 mEq/mL in dextrose solution and infuse over at least 2 hours (maximum rate of administration: 1 mEq/kg/hour).

Vesicant (at concentrations ≥8.4%); ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote (Ref).

Hyaluronidase: **Intradermal or SUBQ:** Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if IV catheter remains in place, administer IV through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution (Ref).

Oral:

Powder: Do not administer when overly full from food or drink. Dissolve ½ teaspoonful of powder completely in 120 mL of water prior to administering.

Tablet: Do not administer when overly full from food or drink. May be swallowed whole or dissolved completely in a glass of water prior to administering. Administration instructions differ by product; refer to product-specific labeling for additional administration instructions.

Infiltration (dental use; Onpharma): Add specified volume of 8.4% sodium bicarbonate directly with lidocaine and epinephrine injection and mix; use immediately after mixing.

Preparation for Administration: Adult

Prevention of contrast-induced nephropathy (off-label use): Remove 154 mL from 1,000 mL bag of D5W; replace with 154 mL of 8.4% sodium bicarbonate; resultant concentration is 154 mEq/L (Ref); more practically, institutions may remove 150 mL from 1,000 mL bag of D5W and replace with 150 mL of 8.4% sodium bicarbonate; resultant concentration is 150 mEq/L.

Neutralizing additive (dental use): Add specified volume of 8.4% sodium bicarbonate directly with lidocaine and epinephrine injection and mix; use immediately after mixing.

Administration: Pediatric

Oral:

Powder for solution (baking soda): Measure dose exactly, mix with adequate amount of water (eg, $^{1}/_{2}$ teaspoon powder with 4 ounces of water), allow to dissolve completely prior to administration.

Infiltration (neutralizing additive, dental use): Add specified volume of 8.4% sodium bicarbonate parenteral solution directly with lidocaine and epinephrine injection and mix; use immediately after mixing.

Parenteral:

Note: Commercially available products: Each mL of 4.2% IV product provides 0.5 mEq/mL of sodium and 0.5 mEq/mL of bicarbonate. Each mL of 8.4% IV product provides 1 mEq/mL of sodium and 1 mEq/mL of bicarbonate.

Direct IV injection:

Neonates and Infants: Administer 0.5 mEq/mL solution; infusion rate depends upon indication, see dosing (Ref).

Children and Adolescents: Administer 1 mEq/mL solution; infusion rate depends upon indication, see dosing; maximum rate: 10 mEq/minute (Ref).

IV infusion: Infusion time variable based upon indication; may be added to IV fluids for infusion when urgent correction is not needed; for metabolic acidosis, infusions over 4 to 8 hours have been suggested (Ref); in neonates, the maximum rate of administration: 1 mEq/kg/hour (Ref).

Vesicant (at concentrations ≥8.4%); ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation. If extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for IV antidote; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote (Ref).

Topical: Paste, compress/dressing: Apply to clean affected area.

Preparation for Administration: Pediatric

Parenteral:

IV:

ASHP recommends standard concentrations of 0.5 mEq/mL and 1 mEq/mL in pediatric patients (Ref).

Neutralizing additive (dental use): Add specified volume of 8.4% sodium bicarbonate directly with lidocaine and epinephrine injection and mix (see manufacturer labeling for additional details); use immediately after mixing.

Topical:

Bath soak: Topical: Dissolve 1 to 2 cups of powder in tub of warm water.

Compress or wet dressing: Topical: Mix powder with water; soak clean, soft cloth in mixture; discard mixture after each use.

Paste: Topical: Mix powder with appropriate amount of water to form a paste.

Vesicant/Extravasation Risk

Vesicant (at concentrations ≥8.4%)

Storage/Stability

Injection: Store at 20°C to 25°C (68°F to 77°F). Protect from freezing.

Oral: Store at 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Antacid: Relief of acid indigestion, heartburn, sour stomach, and upset stomach.

Metabolic acidosis: Management of metabolic acidosis.

Neutralizing additive (IV use): To reduce the incidence of chemical phlebitis and patient discomfort due to vein irritation at or near the infusion site by raising the pH of IV acid solutions.

Neutralizing additive (dental use): Improves onset of analgesia and reduces injection site pain by adjusting lidocaine with epinephrine solution to a more physiologic pH.

Urine alkalinization: Alkalinization agent for urine, including management of specific toxicities (eg, salicylates).

Use: Off-Label: Adult

Cardiac conduction delays due to sodium channel blockade; Contrast-induced nephropathy (prevention); Hyperkalemia, severe/emergent

Medication Safety Issues

Other safety concerns:

Na bicarbonate is an error-prone way of saying sodium bicarbonate (mistaken as no bicarbonate)

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acalabrutinib: Antacids may decrease serum concentrations of Acalabrutinib. Management: Separate administration of acalabrutinib capsules from the administration of any antacid by at least 2 hours in order to minimize the potential for a significant interaction. Acalabrutinib tablets are not expected to interact with antacids. *Risk D: Consider Therapy Modification*

AcetaZOLAMIDE: May increase adverse/toxic effects of Sodium Bicarbonate (Systemic). Specifically, the risk of renal calculus formation may be increased. *Risk C: Monitor*

Amantadine: Alkalinizing Agents may increase serum concentrations of Amantadine. *Risk C: Monitor*

Amphetamines: Alkalinizing Agents may decrease excretion of Amphetamines. Management: Consider alternatives to using amphetamines and alkalinizing agents in combination. If these agents must be used together, patients should be monitored closely for excessive amphetamine effects. *Risk D: Consider Therapy Modification*

Antipsychotic Agents (Phenothiazines): Antacids may decrease absorption of Antipsychotic Agents (Phenothiazines). *Risk C: Monitor*

Atazanavir: Antacids may decrease absorption of Atazanavir. Management: Administer antacids 1 to 2 hours before or 2 hours after atazanavir to minimize the risk of a clinically significant interaction. *Risk D: Consider Therapy Modification*

Belumosudil: Antacids may decrease serum concentrations of Belumosudil. Management: Consider separating administration of belumosudil and antacids by 2 hours and monitor for reduced belumosudil efficacy. *Risk D: Consider Therapy Modification*

Bisacodyl: Antacids may decrease therapeutic effects of Bisacodyl. Antacids may cause the delayed-release bisacodyl tablets to release drug prior to reaching the large intestine. Gastric irritation and/or cramps may occur. Management: Antacids should not be used within 1 hour before bisacodyl administration. *Risk D: Consider Therapy Modification*

Bismuth Subcitrate: Antacids may decrease therapeutic effects of Bismuth Subcitrate. Management: Administer antacids at least 30 minutes before or 30 minutes after bismuth subcitrate administration. *Risk D: Consider Therapy Modification*

Bosutinib: Antacids may decrease serum concentrations of Bosutinib. Management: Administer antacids more than 2 hours before or after bosutinib. *Risk D: Consider Therapy Modification*

Bromperidol: Antacids may decrease absorption of Bromperidol. *Risk C: Monitor*

Calcium Polystyrene Sulfonate: Antacids may increase adverse/toxic effects of Calcium Polystyrene Sulfonate. The combined use of these two agents may result in metabolic alkalosis. Antacids may decrease bradycardic effects of Calcium Polystyrene Sulfonate. *Risk C: Monitor*

Captopril: Antacids may decrease serum concentrations of Captopril. *Risk C: Monitor*

Cefditoren: Antacids may decrease serum concentrations of Cefditoren. Management: Concomitant use of antacids and cefditoren is not recommended. Consider separating the administration of these agents by at least 2 hours. *Risk D: Consider Therapy Modification*

Cefpodoxime: Antacids may decrease serum concentrations of Cefpodoxime. *Risk C: Monitor*

Cefuroxime: Antacids may decrease serum concentrations of Cefuroxime. Management: Administer cefuroxime axetil at least 1 hour before or 2 hours after the administration of short-acting antacids. Taking cefuroxime with food may lessen the magnitude of this interaction. *Risk D: Consider Therapy Modification*

Chloroquine: Antacids may decrease serum concentrations of Chloroquine. Management: Separate the administration of antacids and chloroquine by at least 4 hours to minimize any potential negative impact of antacids on chloroquine bioavailability. *Risk D: Consider Therapy Modification*

Cysteamine (Systemic): Antacids may decrease therapeutic effects of Cysteamine (Systemic). *Risk C: Monitor*

Dabigatran Etexilate: Antacids may decrease serum concentrations of Dabigatran Etexilate. Management: Dabigatran etexilate Canadian product labeling recommends avoiding concomitant use with antacids for 24 hours after surgery. In other situations, administer dabigatran etexilate 2 hours prior to antacids. Monitor clinical response to dabigatran therapy. *Risk D: Consider Therapy Modification*

Dasatinib: Antacids may decrease serum concentrations of Dasatinib. Management: Simultaneous administration of dasatinib and antacids should be avoided. If combined, administer antacids 2 hours before or 2 hours after dasatinib. *Risk D: Consider Therapy Modification*

Defactinib: Antacids may decrease serum concentrations of Defactinib. Antacids may decrease active metabolite exposure of Defactinib. Management: Administer locally acting antacids 2 hours before or after defactinib. *Risk D: Consider Therapy Modification*

DexAMETHasone (Systemic): Antacids may decrease bioavailability of DexAMETHasone (Systemic). *Risk C: Monitor*

Erlotinib: Antacids may decrease serum concentrations of Erlotinib. Management: Separate the administration of erlotinib and any antacid by several hours in order to minimize the risk of a significant interaction. *Risk D: Consider Therapy Modification*

Flecainide: Sodium Bicarbonate (Systemic) may decrease excretion of Flecainide. *Risk C: Monitor*

Fosinopril: Antacids may decrease serum concentrations of Fosinopril. Management: Separate administration of antacids and fosinopril by 2 hours. *Risk D: Consider Therapy Modification*

Gefitinib: Antacids may decrease serum concentrations of Gefitinib. Management: Administer gefitinib at least 6 hours before or 6 hours after administration of an antacid, and closely monitor clinical response to gefitinib. *Risk D: Consider Therapy Modification*

Hyoscyamine: Antacids may decrease serum concentrations of Hyoscyamine. Management: Administer immediate release and sublingual oral hyoscyamine before meals, and antacids after meals, when these agents are given in combination. *Risk D: Consider Therapy Modification*

Iron Preparations: Antacids may decrease absorption of Iron Preparations. Management: No action is likely necessary for the majority of patients who only use antacids intermittently or occasionally. Consider separating doses of oral iron and antacids in patients who require chronic use of both agents and monitor for reduced iron efficacy. *Risk D: Consider Therapy Modification*

Isoniazid: Antacids may decrease absorption of Isoniazid. *Risk C: Monitor*

Itraconazole: Antacids may decrease serum concentrations of Itraconazole. Antacids may increase serum concentrations of Itraconazole. Management: Administer Sporanox brand itraconazole at least 2 hours before or 2 hours after administration of any antacids. Exposure to Tolsura brand itraconazole may be increased by antacids; consider itraconazole dose reduction. *Risk D: Consider Therapy Modification*

Ketoconazole (Systemic): Antacids may decrease serum concentrations of Ketoconazole (Systemic). Management: Administer antacids at least 1 hour prior to, or 2 hours after, ketoconazole. Additionally, administer ketoconazole with an acidic beverage (eg, non-diet cola) and monitor patients closely for signs of inadequate clinical response to ketoconazole. *Risk D: Consider Therapy Modification*

Lanthanum: Antacids may decrease therapeutic effects of Lanthanum. Management: Administer antacid products at least 2 hours before or after lanthanum. *Risk D: Consider Therapy Modification*

Ledipasvir: Antacids may decrease serum concentrations of Ledipasvir. Management: Separate the administration of ledipasvir and antacids by 4 hours. *Risk D: Consider Therapy Modification*

Levoketoconazole: Antacids may decrease absorption of Levoketoconazole. Management: Advise patients to take antacids at least 2 hours after taking levoketoconazole. *Risk D: Consider Therapy Modification*

Levonadifloxacin: Antacids may decrease serum concentrations of Levonadifloxacin. *Risk X: Avoid*

Mecamylamine: Alkalinizing Agents may increase serum concentrations of Mecamylamine. *Risk C: Monitor*

Memantine: Alkalinizing Agents may increase serum concentrations of Memantine. *Risk C: Monitor*

Mesalamine: Antacids may decrease therapeutic effects of Mesalamine. Antacid-mediated increases in gastrointestinal pH may cause the premature release of mesalamine from specific sustained-release mesalamine products. Management: Avoid concurrent administration of antacids with the Apriso brand of mesalamine extended-release capsules. The optimal duration of dose separation is unknown. Other mesalamine products do not contain this interaction warning. *Risk D: Consider Therapy Modification*

Methenamine: Antacids may decrease therapeutic effects of Methenamine. Management: Consider avoiding the concomitant use of medications that alkalinize the urine, such as antacids, and methenamine. Monitor for decreased therapeutic effects of methenamine if used concomitant with antacids. *Risk D: Consider Therapy Modification*

Multivitamins/Minerals (with ADEK, Folate, Iron): Antacids may decrease serum concentrations of Multivitamins/Minerals (with ADEK, Folate, Iron). Specifically, antacids may decrease the absorption of orally administered iron. Management: Separate dosing of oral iron-containing multivitamins and antacids by as much time as possible to minimize impact of this interaction. Monitor for decreased therapeutic efficacy of oral iron preparations during coadministration. *Risk D: Consider Therapy Modification*

Neratinib: Antacids may decrease serum concentrations of Neratinib. Specifically, antacids may reduce neratinib absorption. Management: Separate the administration of neratinib and antacids by giving neratinib at least 3 hours after the antacid. *Risk D: Consider Therapy Modification*

Nilotinib: Antacids may decrease serum concentrations of Nilotinib. Management: Separate the administration of nilotinib and any antacid by at least 2 hours whenever possible in order to minimize the risk of a significant interaction. *Risk D: Consider Therapy Modification*

Nirogacestat: Antacids may decrease serum concentrations of Nirogacestat. Management: If acid-reducing therapy is required, separate nirogacestat administration from antacids by 2 hours. *Risk D: Consider Therapy Modification*

Octreotide: Antacids may decrease serum concentrations of Octreotide. *Risk C: Monitor*

PAZOPanib: Antacids may decrease serum concentrations of PAZOPanib. Management: Avoid the use of antacids in combination with pazopanib whenever possible. Separate doses by several hours if antacid treatment is considered necessary. The impact of dose separation has not been investigated. *Risk D: Consider Therapy Modification*

Pexidartinib: Antacids may decrease serum concentrations of Pexidartinib. Management: Administer pexidartinib at least 2 hours before or 2 hours after antacids. *Risk D: Consider Therapy Modification*

Phosphate Supplements: Antacids may decrease absorption of Phosphate Supplements. Management: This applies only to oral phosphate administration. Separating administration of oral phosphate supplements from antacid administration by as long as possible may minimize the interaction. *Risk D: Consider Therapy Modification*

Potassium Phosphate: Antacids may decrease serum concentrations of Potassium Phosphate. Management: Consider separating administration of antacids and oral potassium phosphate by at least 2 hours to decrease risk of a significant interaction. *Risk D: Consider Therapy Modification*

PredniSONE: Antacids may decrease bioavailability of PredniSONE. *Risk C: Monitor*

QuiNIDine: Alkalinizing Agents may increase serum concentrations of QuiNIDine. *Risk C: Monitor*

QuiNINE: Alkalinizing Agents may increase serum concentrations of QuiNINE. *Risk C: Monitor*

Rilpivirine: Antacids may decrease serum concentrations of Rilpivirine. Management: Administer antacids at least 2 hours before or 4 hours after oral rilpivirine when used with most rilpivirine products. However, administer antacids at least 6 hours before or 4 hours after the rilpivirine/dolutegravir combination product. *Risk D: Consider Therapy Modification*

Rilzabrutinib: Antacids may decrease serum concentrations of Rilzabrutinib. Management: Administer antacids at least 2 hours after rilzabrutinib. *Risk D: Consider Therapy Modification*

Riociguat: Antacids may decrease serum concentrations of Riociguat. Management: Separate the administration of antacids and riociguat by at least 1 hour in order to minimize any potential interaction. Monitor clinical response to riociguat more closely in patients using this combination. *Risk D: Consider Therapy Modification*

Rosuvastatin: Antacids may decrease serum concentrations of Rosuvastatin. *Risk C: Monitor*

Selpercatinib: Antacids may decrease serum concentrations of Selpercatinib. Management: Coadministration of selpercatinib and antacids should be avoided. If coadministration cannot be avoided, selpercatinib should be administered 2 hours before or 2 hours after antacids. *Risk D: Consider Therapy Modification*

Sodium Polystyrene Sulfonate: Antacids may increase adverse/toxic effects of Sodium Polystyrene Sulfonate. Specifically, the risk of metabolic alkalosis may be increased. Antacids may decrease therapeutic effects of Sodium Polystyrene Sulfonate. *Risk C: Monitor*

Sotalol: Antacids may decrease serum concentrations of Sotalol. Management: Avoid simultaneous administration of sotalol and antacids. Administer antacids 2 hours after sotalol. *Risk D: Consider Therapy Modification*

Sotorasib: Antacids may decrease serum concentrations of Sotorasib. Management: Avoid use with PPIs, PCABs, or H2RAs. If use of a gastric acid suppressing medication is needed, use antacids and administer sotorasib 4 hours before or 10 hours after oral antacid administration. *Risk D: Consider Therapy Modification*

Sucralfate: Antacids may decrease therapeutic effects of Sucralfate. Management: Consider separating the administration of antacids and sucralfate by at least 30 minutes. *Risk D: Consider Therapy Modification*

Sulpiride: Antacids may decrease serum concentrations of Sulpiride. Management: Separate administration of antacids and sulpiride by at least 2 hours in order to minimize the impact of antacids on sulpiride absorption. *Risk D: Consider Therapy Modification*

Taletrectinib: Antacids may decrease serum concentrations of Taletrectinib. Management: Administer antacids 2 hours before or 2 hours after taletrectinib. *Risk D: Consider Therapy Modification*

Tetracyclines: Sodium Bicarbonate (Systemic) may decrease serum concentrations of Tetracyclines. *Risk C: Monitor*

Thyroid Products: Antacids may decrease absorption of Thyroid Products. *Risk C: Monitor*

Velpatasvir: Antacids may decrease serum concentrations of Velpatasvir. Management: Separate administration of velpatasvir and antacids by 4 hours. *Risk D: Consider Therapy Modification*

Ziftomenib: Antacids may decrease serum concentrations of Ziftomenib. Management: Avoid coadministration of antacids with ziftomenib. If coadministration cannot be avoided, administer ziftomenib 2 hours before or 2 hours after the antacid. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Antacids containing sodium bicarbonate should not be used during pregnancy due to their potential to cause metabolic alkalosis and fluid overload in the mother and fetus (Body 2016; Dağlı 2017; Gomes 2018; Thélin 2020).

Breastfeeding Considerations

Sodium is found in breast milk (IOM 2004).

Dietary Considerations

Some products may contain sodium. Oral product should be taken 1 to 3 hours after meals.

Monitoring Parameters

Serum electrolytes (including bicarbonate, potassium, sodium, calcium), urinary pH, arterial blood gases (if indicated).

Mechanism of Action

Dissociates to provide bicarbonate ion which neutralizes hydrogen ion concentration and raises blood and urinary pH

Neutralizing additive (dental use): Increases pH of lidocaine and epinephrine solution to improve tolerability and increase tissue uptake

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Oral: 15 minutes; IV: Rapid

Duration: Oral: 1 to 3 hours; IV: 8 to 10 minutes

Absorption: Oral: Well absorbed

Excretion: Urine (<1%)

Patient Counseling Points

(For additional information see "Sodium bicarbonate: Patient drug information")

What is this drug used for?

Tablets:

• It is used to treat heartburn and upset stomach.

Injection:

• It is used to lower acid levels in the body.

• It is used to replace bicarbonate loss caused by severe diarrhea.

All products:

• It may be given to you for other reasons. Talk with the doctor.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

• Twitching

• Muscle stiffness

• Muscle spasms

• Feeling irritable

• Shortness of breath, a big weight gain, or swelling in the arms or legs

• Redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Sodium Bicarbonate bp;
- (AR) Argentina: Bicarbonato | Bicarbonato de sodio | Bicarbonato de sodio ewe | Bicarbonato de sodio I.B.C. | Bicarbonato de sodio puler | Bicarbonato sodio | Solucion molar bicarbonato de sodio tecsolpar | Solucion molar de bicarbonato de sodio | Solucion molar de bicarbonato de sodio i.v.;
- (AT) Austria: Bullrich salz | Natrium bikarbonat fresenius kabi | Nephrotrans;
- (BD) Bangladesh: Sodibar | Sodicarb | Sodinate;
- (BE) Belgium: Braun solute | Na bicarbonate baxter | Natribic | Sodium bicarbonate vascumed povite;
- (BG) Bulgaria: Natrium hydrocarbonicum | Sodium bicarbonate braun 8,4%;
- (BR) Brazil: Bibag | Bicarbon | Bicarbonato de sodio | Bicarbonato de sodio be better | Bicarbonato de sodio bem basico | Bicarbonato de sodio farmax | Bicarbonato de sodio lifar | Bicarbonato de sodio needs | Bicarbonato de sodio nexter | Bicarbonato de sodio sidney oliveira | Bicarbonato de sodio v.help | Bicarbonato sodio | Drogasil bicarbonato de sodio | Farmacart | Polyfarma bicarbonato de sodio | Sol cart b | Solucao de bicarbonato de sodio halex istar;
- (CH) Switzerland: Alkala | Natrium Bicarbonat B. Braun | Natrium bicarbonat braun | Natrium bicarbonicum bichsel | Natrium bicarbonicum sintetica;
- (CL) Chile: Bicarbonato de sodio;
- (CO) Colombia: Bibag | Bicarbonato de sodio | Bicarbonato sodio;
- (CR) Costa Rica: Bicarbonato de sodio;
- (CZ) Czech Republic: Alkaligen | Natriumhydrogenca;
- (DE) Germany: Bicanorm | Bullrich salz | Min-i-jet natrium hydroxid | Natriumhydrogencarbonat B. Braun | Natriumhydrogencarbonat Baxter | Natriumhydrogencarbonat DeltaSelect | Natriumhydrogencarbonat koehler | Nephrotrans;
- (EC) Ecuador: Bicarbonato de sodio | Bicarbonato sodico | Sobisis;
- (EE) Estonia: Natrii hydrocarb;
- (ES) Spain: Apir Bicarbonato sodico | Bicarbonato | Bicarbonato brum | Bicarbonato cinfa | Bicarbonato de sodio orravan | Bicarbonato de sodio pege | Bicarbonato pyre | Bicarbonato sod Bieffe | Bicarbonato sod pg | Bicarbonato sodico | Bicarbonato sodico Braun | Bicarbonato sodico Mein | Bicarbonato sodico Viviar | Bicarbonato viviar | Venofusin bicarbonato sodico | Viaflo bicarbonato sodico;
- (FI) Finland: Bicart | Na bicarb isot le | Na bicarb isot mpl | Natr bicarb medip | Natr bicarb ro | Natrium bicarbonate orion | Natriumbicarbonate Braun;
- (FR) France: Bicafres | Bicarbonate de sodium aguettant | Bicarbonate de sodium B.Braun | Bicarbonate de sodium baxter | Bicarbonate de sodium biosedra | Bicarbonate de sodium cooper | Bicarbonate de Sodium Gifrer | Bicarbonate de sodium lavoisier | Bicarbonate de sodium renaudin;
- (GB) United Kingdom: Nephrotrans | Polyfusor sodium bicarbonate;
- (HK) Hong Kong: Antacimin | Sodamint;
- (HR) Croatia: Natrijev hidrogenkarbonat medice;
- (HU) Hungary: Alkaligen | Natrium hydrogen carbonicum PM | Natrium hydrogencarbonate;
- (ID) Indonesia: Bicarbic natricus | Bicarbonat | Meylon | Natrium bicarbonat;
- (IE) Ireland: Braun sodium bicarbonate;

- (IN) India: Acidose | Acidose ds | Auxisoda | Bicarcid | Diosis | Nabico3 | Natricarb | Nocidos | Nodosis | Rencarb | Sarpix | Sobicid ds | Sobinix | Sobisis | Sobiup | Socadium | Sodabicarb | Sodafix ec | Sodakindle | Sodanet | Sodanet ds | Sodanet gst | Soddy | Sodisave | Sodonish ds;
- (IT) Italy: Bicart | Sodico bicarbonato | Sodio Bicarbon | Sodio bicarbonato | Sodio bicarbonato galenica senese | Sodio Bicarbonato Marco Viti Farmaceutici | Sol Bicarbon.Sodio | Sol Bilanciata | Sol Bilanciata R+K;
- (JP) Japan: Berin | Jusonin | Jutamin | Jutamin pl | Meylon | Prebinate | Sod.bicarb.bioferm | Sod.bicarb.sankei | Sodium bicarbonate fuso | Sodium Bicarbonate Hikari | Sodium bicarbonate jp | Sodium bicarbonate merck hoei | Sodium bicarbonate nipro | Sodium bicarbonate ns | Sodium bicarbonate pl fuso | Sodium Bicarbonate Towa | Tansonin;
- (KE) Kenya: Sobisis forte | Sodium bicarbonate bp (sodavita);
- (KR) Korea, Republic of: Bi | Bibag | Bicart | Biocapgi | Bivontase | Cdsb | Diacart | Esbi | Latna | Solcart b | Solcartbi | Tasna;
- (LB) Lebanon: Bicanorm;
- (LT) Lithuania: Natrium hydrocarbonas;
- (LV) Latvia: Natrium hydrocarbonat;
- (MA) Morocco: Bicarbonate NA;
- (MX) Mexico: Betsol z | Betuciaven | Bicarbonat sod | Bicarnat | Tecsobin;
- (MY) Malaysia: Solcart b;
- (NL) Netherlands: Natriumhydrocarbonate;
- (NO) Norway: Natriumhydrogencarbonat koehler | Natriumhydrogenkarbonat | Nephrotrans | Sodio bicarbonato;
- (PA) Panama: Bibag | Bicar nat | Bicarbonato de sodio | Bicart | Bipodial | Sol cart b;
- (PE) Peru: Bicarbonato de sodio;
- (PH) Philippines: Bicargen | Soda mint | Sodium bicarbonate atlantic | Solunate | Supracid;
- (PK) Pakistan: Pcarb | Sidocarb | Sodamint;
- (PL) Poland: Alcaneff | Binat | Natrium bicarbonicum;
- (PT) Portugal: Basocare | Bibag | Bicarbonato de sodio | Bicarbonato sodio Labesfal;
- (PY) Paraguay: Bicarbonato de sodio catedral | Renacarb;
- (QA) Qatar: BiCarb-600 | Molar Sodium Bicarbonate;
- (RO) Romania: Bicarbonat de sodiu | Bicarbonat de sodiu infomed;
- (RU) Russian Federation: Natrii hydrocarb. | Sodium hydrocarbonate | Sodium hydrocarbonate eskom | Vitar soda;
- (SA) Saudi Arabia: Nabica;
- (SE) Sweden: Bicart | Natriumbikarbonat evolan | Natriumbikarbonat Fresenius Kabi | Natriumbikarbonat recip | Natriumbikoarbonat Recip;
- (SK) Slovakia: Natrium hydrocarbonicum;
- (SV) El Salvador: Bicarnat;
- (TH) Thailand: Arc mint | Arcsoda | Gasotab | Sodamate | Sodamint | Sodamint frx;
- (TN) Tunisia: Bicarbonate de sodium baxter | Serum bicarbonate;
- (TR) Turkey: Karbidoz | Molar sodium bicarbonat | Molar sodium bicarbonate osel | Molar sodyum bikarbonat osel | Norm asidoz | Sodium bicarbonate molar | Sodyum bikarbonat;
- (TW) Taiwan: Bicarbona | Bicart | Compound sodium bicarbonate | Meylon | Naturalyte 4000 | Naturalyte 9000 | Rolikan | Sodium Bicarbonate Compound | Stomach;
- (UA) Ukraine: Natrium bicarbonat | Natrium hydrocarbonate | Natrium hydrocarbonicum | Soda bufer;
- (UG) Uganda: Sobisis | Sobisis forte;
- (UY) Uruguay: Bicarbonato sodico | Bicarbonato sodio | Bicarfu | Farmabag | Solucion de bicarbonato cobe | Suero Bicarbonatado | Suero bicarbonatado 1/6 molar;
- (VE) Venezuela, Bolivarian Republic of: Bicarbonato de sodio;
- (VN) Viet Nam: Nabifar;
- (ZM) Zambia: Intramed sodium bicarbonate;
- (ZW) Zimbabwe: Sol cart b

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